

Metamath Predicate Calculus

Phase I: breadth-first mastery through specialization

A dependency-ordered problem set for the modified Metamath Explorer presentation with equality and one primitive unary predicate P . The terminal milestone is the unrestricted specialization theorem $\vdash (\forall x\phi \rightarrow \phi)$.

Design contract

- The previously mastered classical propositional calculus is available as a lemma library; no purely propositional theorem appears as an exercise.
- Every exercise may use the axioms, definitions, inference rules, and only earlier numbered exercises.
- Formula metavariables ϕ, ψ, χ range over wffs. Individual metavariables x, y, z range over individual variables. Metavariable substitution is simultaneous.
- $\$d\ x\ \phi\ \$$ means that every expression substituted for x is variable-disjoint from every expression substituted for ϕ . A displayed $\$d$ condition is part of the exercise.
- An inference-form exercise licenses the resulting derived rule after it has been proved. Its displayed premises are theorem hypotheses, not temporary object-language assumptions.
- Phase I uses ax-1 through ax-7, ax-P, ax-12, modus ponens, generalization, and df-ex. Ax-10, ax-11, and ax-13 belong to the full system but are behind the Phase I dependency firewall.

Formal base

Primitive formation rules

- φ, ψ, χ are wffs; x, y, z are individual variables.
- If φ and ψ are wffs, then $\neg\varphi$ and $(\varphi \rightarrow \psi)$ are wffs.
- If φ is a wff and x is an individual variable, then $\forall x\varphi$ is a wff.
- If x and y are individual variables, then $x=y$ is a wff.
- If x is an individual variable, then $P(x)$ is a wff.
- The connectives $\wedge, \vee$, and $\leftrightarrow$ are the already-mastered propositional definitions.
- Existence is defined by df-ex: $\vdash (\exists x\varphi \leftrightarrow \neg\forall x\neg\varphi)$.

Rules

ax-mp: from $\vdash \varphi$ and $\vdash (\varphi \rightarrow \psi)$, infer $\vdash \psi$.

ax-gen: from $\vdash \varphi$, infer $\vdash \forall x\varphi$.

Propositional axioms

ax-1 $\vdash (\varphi \rightarrow (\psi \rightarrow \varphi))$

ax-2 $\vdash ((\varphi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \chi)))$

ax-3 $\vdash ((\neg\varphi \rightarrow \neg\psi) \rightarrow (\psi \rightarrow \varphi))$

Predicate and equality axioms

ax-4 $\vdash (\forall x(\varphi \rightarrow \psi) \rightarrow (\forall x\varphi \rightarrow \forall x\psi))$

ax-5 $\vdash (\varphi \rightarrow \forall x\varphi) \quad [\$d \ x \ \varphi \ \$.]$

ax-6 $\vdash \neg\forall x\neg(x=y)$

ax-7 $\vdash (x=y \rightarrow (x=z \rightarrow y=z))$

ax-P $\vdash (x=y \rightarrow (P(x) \rightarrow P(y)))$

ax-10 $\vdash (\neg\forall x\varphi \rightarrow \forall x\neg\forall x\varphi)$

ax-11 $\vdash (\forall x\forall y\varphi \rightarrow \forall y\forall x\varphi)$

ax-12 $\vdash (x=y \rightarrow (\forall y\varphi \rightarrow \forall x(x=y \rightarrow \varphi)))$

ax-13 $\vdash (\neg x=y \rightarrow (y=z \rightarrow \forall x(y=z)))$

Phase I firewall: ax-10, ax-11, and ax-13 are listed so the complete system remains explicit, but no exercise below requires them. Do not use them in this phase.

Exercises

I. Generalization and quantified propositional bases

Begin with one-step uses of generalization. Every target contains genuinely predicate-logical syntax, even when its matrix is a previously mastered propositional theorem.

1. $\vdash \forall x(\varphi \rightarrow \varphi)$
2. $\vdash \forall y(\varphi \rightarrow \varphi)$
3. $\vdash \forall x\forall y(\varphi \rightarrow \varphi)$
4. $\vdash \forall y\forall x(\varphi \rightarrow \varphi)$
5. $\vdash \forall x(P(x) \rightarrow P(x))$
6. $\vdash \forall y(P(x) \rightarrow P(x))$
7. $\vdash \forall x(x=y \rightarrow x=y)$
8. $\vdash \forall z(x=y \rightarrow x=y)$
9. $\vdash \forall x(\varphi \vee \neg\varphi)$
10. $\vdash \forall x\neg(\varphi \wedge \neg\varphi)$
11. $\vdash \forall x(\varphi \leftrightarrow \varphi)$
12. $\vdash \forall x((\varphi \wedge \psi) \rightarrow \varphi)$
13. $\vdash \forall x((\varphi \wedge \psi) \rightarrow \psi)$
14. $\vdash \forall x(\varphi \rightarrow (\psi \rightarrow \varphi))$
15. $\vdash \forall x((\varphi \rightarrow \psi) \rightarrow (\neg\psi \rightarrow \neg\varphi))$

II. Universal transport

Build the standard lifting machinery generated by ax-4. The two inference-form exercises are intended to become reusable proof tools.

16. $\vdash (\forall x(\varphi \rightarrow \psi) \rightarrow (\forall x\varphi \rightarrow \forall x\psi))$
17. $\vdash (\forall x(P(x) \rightarrow \neg\neg P(x)) \rightarrow (\forall xP(x) \rightarrow \forall x\neg\neg P(x)))$
18. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\forall x\varphi \rightarrow \forall x\psi)$.
19. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\forall x\forall y\varphi \rightarrow \forall x\forall y\psi)$.
20. $\vdash (\forall x\varphi \rightarrow \forall x\neg\neg\varphi)$
21. $\vdash (\forall x\neg\neg\varphi \rightarrow \forall x\varphi)$
22. $\vdash (\forall x\neg\neg\varphi \leftrightarrow \forall x\varphi)$
23. $\vdash (\forall x(\varphi \wedge \psi) \rightarrow \forall x\varphi)$
24. $\vdash (\forall x(\varphi \wedge \psi) \rightarrow \forall x\psi)$
25. $\vdash (\forall x\varphi \rightarrow (\forall x\psi \rightarrow \forall x(\varphi \wedge \psi)))$
26. $\vdash (\forall x(\varphi \wedge \psi) \leftrightarrow (\forall x\varphi \wedge \forall x\psi))$
27. $\vdash (\forall x\varphi \rightarrow \forall x(\varphi \vee \psi))$

28. $\vdash (\forall x \psi \rightarrow \forall x(\varphi \vee \psi))$
29. $\vdash ((\forall x \varphi \vee \forall x \psi) \rightarrow \forall x(\varphi \vee \psi))$
30. $\vdash (\forall x(\varphi \rightarrow \psi) \rightarrow (\forall x \neg \psi \rightarrow \forall x \neg \varphi))$
31. $\vdash (\forall x(\varphi \leftrightarrow \psi) \rightarrow (\forall x \varphi \leftrightarrow \forall x \psi))$
32. $\vdash (\forall x \varphi \rightarrow \forall x(\psi \rightarrow \varphi))$
33. $\vdash (\forall x(\varphi \rightarrow \psi) \rightarrow (\forall x(\psi \rightarrow \chi) \rightarrow (\forall x \varphi \rightarrow \forall x \chi)))$
34. $\vdash (\forall x(\varphi \wedge \psi) \rightarrow \forall x(\psi \wedge \varphi))$
35. $\vdash (\forall x(\varphi \vee \psi) \rightarrow \forall x(\psi \vee \varphi))$
36. $\vdash (\forall x P(x) \leftrightarrow \forall x \neg \neg P(x))$
37. $\vdash \forall x(P(x) \vee \neg P(x))$
38. $\vdash \forall x \neg(P(x) \wedge \neg P(x))$

III. Existential definition and quantifier duality

Expand $\exists$ only through df-ex, then transport propositional equivalences through $\forall$. Work both directions separately before packaging biconditionals.

39. $\vdash (\exists x \varphi \rightarrow \neg \forall x \neg \varphi)$
40. $\vdash (\neg \forall x \neg \varphi \rightarrow \exists x \varphi)$
41. $\vdash (\forall x \neg \varphi \leftrightarrow \neg \exists x \varphi)$
42. $\vdash (\forall x \neg \varphi \rightarrow \neg \exists x \varphi)$
43. $\vdash (\neg \exists x \varphi \rightarrow \forall x \neg \varphi)$
44. $\vdash (\forall x \varphi \leftrightarrow \neg \exists x \neg \varphi)$
45. $\vdash (\forall x \varphi \rightarrow \neg \exists x \neg \varphi)$
46. $\vdash (\neg \exists x \neg \varphi \rightarrow \forall x \varphi)$
47. $\vdash (\exists x \neg \varphi \leftrightarrow \neg \forall x \varphi)$
48. $\vdash (\exists x \neg \varphi \rightarrow \neg \forall x \varphi)$
49. $\vdash (\neg \forall x \varphi \rightarrow \exists x \neg \varphi)$
50. $\vdash (\neg \exists x \varphi \leftrightarrow \forall x \neg \varphi)$
51. $\vdash (\neg \exists x \varphi \rightarrow \forall x \neg \varphi)$
52. $\vdash (\forall x \neg \varphi \rightarrow \neg \exists x \varphi)$
53. $\vdash (\neg \forall x \varphi \leftrightarrow \exists x \neg \varphi)$
54. $\vdash (\neg \forall x \varphi \rightarrow \exists x \neg \varphi)$
55. $\vdash (\exists x \neg \varphi \rightarrow \neg \forall x \varphi)$
56. $\vdash (\exists x P(x) \leftrightarrow \neg \forall x \neg P(x))$
57. $\vdash (\forall x P(x) \leftrightarrow \neg \exists x \neg P(x))$
58. $\vdash (\neg \exists x P(x) \leftrightarrow \forall x \neg P(x))$

IV. Existential transport

Derive monotonicity for $\exists$ from the universal machinery and duality. Keep the generic inference forms available thereafter.

59. $\vdash (\forall x(\varphi \rightarrow \psi) \rightarrow (\exists x\varphi \rightarrow \exists x\psi))$
60. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\exists x\varphi \rightarrow \exists x\psi)$.
61. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\exists x\exists y\varphi \rightarrow \exists x\exists y\psi)$.
62. $\vdash (\exists x(\varphi \wedge \psi) \rightarrow \exists x\varphi)$
63. $\vdash (\exists x(\varphi \wedge \psi) \rightarrow \exists x\psi)$
64. $\vdash (\exists x\varphi \rightarrow \exists x(\varphi \vee \psi))$
65. $\vdash (\exists x\psi \rightarrow \exists x(\varphi \vee \psi))$
66. $\vdash (\exists x\varphi \rightarrow \exists x\neg\neg\varphi)$
67. $\vdash (\exists x\neg\neg\varphi \rightarrow \exists x\varphi)$
68. $\vdash (\exists x\neg\neg\varphi \leftrightarrow \exists x\varphi)$
69. From $\vdash (\varphi \leftrightarrow \psi)$, infer $\vdash (\exists x\varphi \leftrightarrow \exists x\psi)$.
70. $\vdash (\forall x(\varphi \leftrightarrow \psi) \rightarrow (\exists x\varphi \leftrightarrow \exists x\psi))$
71. $\vdash (\exists x(\varphi \vee \psi) \rightarrow (\exists x\varphi \vee \exists x\psi))$
72. $\vdash ((\exists x\varphi \vee \exists x\psi) \rightarrow \exists x(\varphi \vee \psi))$
73. $\vdash (\exists x(\varphi \vee \psi) \leftrightarrow (\exists x\varphi \vee \exists x\psi))$
74. $\vdash (\exists x(\varphi \wedge \psi) \rightarrow (\exists x\varphi \wedge \exists x\psi))$
75. $\vdash \neg\exists x(\varphi \wedge \neg\varphi)$
76. $\vdash \neg\exists x(P(x) \wedge \neg P(x))$
77. $\vdash \neg\exists x(P(x) \leftrightarrow \neg P(x))$
78. $\vdash (\exists x(P(x) \wedge \varphi) \rightarrow \exists xP(x))$
79. $\vdash (\exists x(P(x) \wedge \varphi) \rightarrow \exists x\varphi)$
80. $\vdash (\exists xP(x) \rightarrow \exists x(P(x) \vee \varphi))$

V. Distinct variables, vacuity, and existential elimination

This is the missing bridge in the original set. Treat each $\$d$ condition as part of the problem. The inference forms are metatheorems about derivations, not extra object-language axioms.

81. $\vdash (\varphi \rightarrow \forall x\varphi)$
Condition: $\$d \times \varphi \ \$$.
82. $\vdash (\neg\varphi \rightarrow \forall x\neg\varphi)$
Condition: $\$d \times \varphi \ \$$.
83. $\vdash (P(y) \rightarrow \forall xP(y))$
Condition: $\$d \times y \ \$$.

84. $\vdash (y=z \rightarrow \forall x(y=z))$

Condition: $\$d \times y \$$ and $\$d \times z \$$.

85. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\varphi \rightarrow \forall x\psi)$.

Condition: $\$d \times \varphi \$$.

86. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\varphi \rightarrow \forall x\forall y\psi)$.

Condition: $\$d \times \varphi \$$ and $\$d \times y \varphi \$$.

87. $\vdash (\varphi \rightarrow \forall x(\psi \rightarrow \varphi))$

Condition: $\$d \times \varphi \$$.

88. $\vdash (\varphi \rightarrow \forall x(\varphi \vee \psi))$

Condition: $\$d \times \varphi \$$.

89. $\vdash ((\varphi \wedge \psi) \rightarrow \forall x\varphi)$

Condition: $\$d \times \varphi \$$ and $\$d \times \psi \$$.

90. $\vdash (\exists x\varphi \rightarrow \varphi)$

Condition: $\$d \times \varphi \$$.

91. $\vdash (\exists x\neg\varphi \rightarrow \neg\varphi)$

Condition: $\$d \times \varphi \$$.

92. $\vdash (\exists x\varphi \rightarrow \forall x\varphi)$

Condition: $\$d \times \varphi \$$.

93. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\exists x\varphi \rightarrow \psi)$.

Condition: $\$d \times \psi \$$.

94. From $\vdash \exists x\varphi$ and $\vdash (\varphi \rightarrow \psi)$, infer $\vdash \psi$.

Condition: $\$d \times \psi \$$.

95. From $\vdash (\chi \rightarrow (\varphi \rightarrow \psi))$, infer $\vdash (\chi \rightarrow (\exists x\varphi \rightarrow \psi))$.

Condition: $\$d \times \chi \$$ and $\$d \times \psi \$$.

96. From $\vdash (\varphi \rightarrow \psi)$, infer $\vdash (\exists x\exists y\varphi \rightarrow \psi)$.

Condition: $\$d \times \psi \$$ and $\$d \times y \psi \$$.

97. $\vdash (\exists x(P(x) \wedge \varphi) \rightarrow \varphi)$

Condition: $\$d \times \varphi \$$.

98. $\vdash (\exists x(\varphi \wedge P(x)) \rightarrow \varphi)$

Condition: $\$d \times \varphi \$$.

99. $\vdash (\exists x(x=y \wedge \varphi) \rightarrow \varphi)$

Condition: $\$d \times \varphi \$$.

100. $\vdash (\exists x(\varphi \wedge x=y) \rightarrow \varphi)$

Condition: $\$d \times \varphi \$$.

101. From $\vdash (P(x) \rightarrow \varphi)$, infer $\vdash (\exists xP(x) \rightarrow \varphi)$.

Condition: $\$d \times \varphi \$$.

102. From $\vdash \exists xP(x)$ and $\vdash (P(x) \rightarrow \varphi)$, infer $\vdash \varphi$.

Condition: $\$d \times \varphi \$$.

$$103. \vdash (\exists x \exists y ((x=y) \wedge \varphi) \rightarrow \varphi)$$

Condition: $\$d \ x \ \varphi \ \$$ and $\$d \ y \ \varphi \ \$$.

VI. Nonempty domains and weak specialization

Introduce ax-6 only after the empty-domain-safe quantifier machinery is established. First obtain witnesses; then prove that a universal statement has an existential instance; only then remove vacuous quantifiers.

$$104. \vdash \neg \forall x \neg (x=y)$$

$$105. \vdash \exists x (x=y)$$

$$106. \vdash \exists y (y=x)$$

$$107. \vdash \exists x (x=x)$$

$$108. \vdash \forall y \exists x (x=y)$$

$$109. \vdash \forall x \exists y (y=x)$$

$$110. \text{ From } \vdash \varphi, \text{ infer } \vdash \exists x \varphi.$$

$$111. \vdash \exists x (\varphi \rightarrow \varphi)$$

$$112. \vdash \exists x (P(x) \vee \neg P(x))$$

$$113. \vdash \exists x \neg (P(x) \wedge \neg P(x))$$

$$114. \vdash (\forall x \varphi \rightarrow \forall x (x=y \rightarrow \varphi))$$

$$115. \vdash (\forall x (x=y \rightarrow \varphi) \rightarrow (\exists x (x=y) \rightarrow \exists x \varphi))$$

$$116. \vdash (\forall x \varphi \rightarrow (\exists x (x=y) \rightarrow \exists x \varphi))$$

$$117. \vdash (\forall x \varphi \rightarrow \exists x \varphi)$$

$$118. \vdash (\forall x \varphi \rightarrow \varphi)$$

Condition: $\$d \ x \ \varphi \ \$$.

$$119. \vdash (\forall x \varphi \leftrightarrow \varphi)$$

Condition: $\$d \ x \ \varphi \ \$$.

$$120. \vdash (\varphi \rightarrow \exists x \varphi)$$

Condition: $\$d \ x \ \varphi \ \$$.

$$121. \vdash (\exists x \varphi \leftrightarrow \varphi)$$

Condition: $\$d \ x \ \varphi \ \$$.

$$122. \vdash (\forall x \varphi \leftrightarrow \exists x \varphi)$$

Condition: $\$d \ x \ \varphi \ \$$.

VII. Equality and the ax-7 degeneracies

All equality-pattern substitutions are made explicit before reflexivity. This prevents the contraction and Rule-C steps from being hidden inside a single jump.

$$123. \vdash (x=y \rightarrow (x=z \rightarrow y=z))$$

$$124. \vdash (x=x \rightarrow (x=z \rightarrow x=z))$$

$$125. \vdash (x=y \rightarrow (x=x \rightarrow y=x))$$

$$126. \vdash (x=y \rightarrow (x=y \rightarrow y=y))$$

127. $\vdash (x=x \rightarrow (x=x \rightarrow x=x))$

128. $\vdash (y=x \rightarrow (y=x \rightarrow x=x))$

129. $\vdash (y=x \rightarrow x=x)$

130. $\vdash (\exists y(y=x) \rightarrow x=x)$

Condition: $\$d \ x \ y \ \$$.

131. $\vdash x=x$

132. $\vdash \forall x(x=x)$

133. $\vdash (x=y \rightarrow y=x)$

134. $\vdash (x=y \leftrightarrow y=x)$

135. $\vdash \exists x(y=x)$

136. $\vdash \forall y \exists x(y=x)$

137. $\vdash (x=y \rightarrow (y=z \rightarrow x=z))$

138. $\vdash (x=y \rightarrow (z=x \rightarrow z=y))$

139. $\vdash (x=y \rightarrow (x=z \leftrightarrow y=z))$

140. $\vdash (x=y \rightarrow (z=x \leftrightarrow z=y))$

141. $\vdash ((x=y \wedge y=z) \rightarrow x=z)$

142. $\vdash ((x=z \wedge y=z) \rightarrow x=y)$

143. $\vdash (x=y \rightarrow (P(x) \rightarrow P(y)))$

144. $\vdash (x=y \rightarrow (P(y) \rightarrow P(x)))$

145. $\vdash (x=y \rightarrow (P(x) \leftrightarrow P(y)))$

146. $\vdash (x=y \rightarrow (\neg P(x) \leftrightarrow \neg P(y)))$

147. $\vdash ((x=y \wedge P(x)) \rightarrow P(y))$

148. $\vdash ((x=y \wedge P(y)) \rightarrow P(x))$

149. $\vdash \forall x \forall y (x=y \rightarrow (P(x) \leftrightarrow P(y)))$

VIII. Ax-12, unrestricted existential introduction, and specialization

The final chain is deliberately granular. Do not compress exercises 158-165 into one library theorem: these are the steps that explain why full specialization is available.

150. $\vdash (x=y \rightarrow (\forall y \varphi \rightarrow \forall x(x=y \rightarrow \varphi)))$

151. $\vdash (y=z \rightarrow (\forall z \varphi \rightarrow \forall y(y=z \rightarrow \varphi)))$

152. $\vdash (\varphi \rightarrow \forall y \varphi)$

Condition: $\$d \ y \ \varphi \ \$$.

153. $\vdash (x=y \rightarrow (\varphi \rightarrow \forall x(x=y \rightarrow \varphi)))$

Condition: $\$d \ x \ y \ \$$ and $\$d \ y \ \varphi \ \$$.

$$154. \vdash (x=y \rightarrow (P(x) \rightarrow \forall x(x=y \rightarrow P(x))))$$

Condition: $\$d \ x \ y \ \$$.

$$155. \vdash (\forall x(x=y \rightarrow \varphi) \rightarrow (\exists x(x=y) \rightarrow \exists x\varphi))$$

$$156. \vdash (\forall x(x=y \rightarrow \varphi) \rightarrow \exists x\varphi)$$

$$157. \vdash (\forall x(x=y \rightarrow P(x)) \rightarrow \exists xP(x))$$

$$158. \vdash (x=y \rightarrow (\varphi \rightarrow \exists x\varphi))$$

Condition: $\$d \ x \ y \ \$$ and $\$d \ y \ \varphi \ \$$.

$$159. \vdash (\exists y(x=y) \rightarrow (\varphi \rightarrow \exists x\varphi))$$

Condition: $\$d \ x \ y \ \$$ and $\$d \ y \ \varphi \ \$$.

$$160. \vdash \exists y(x=y)$$

$$161. \vdash (\varphi \rightarrow \exists x\varphi)$$

$$162. \vdash (\neg\varphi \rightarrow \exists x\neg\varphi)$$

$$163. \vdash (\neg\exists x\neg\varphi \rightarrow \varphi)$$

$$164. \vdash (\forall x\varphi \rightarrow \neg\exists x\neg\varphi)$$

$$165. \vdash (\forall x\varphi \rightarrow \varphi)$$

PHASE I MILESTONE REACHED • Exercise 165: $\vdash (\forall x\varphi \rightarrow \varphi)$